Temos ax^2+bx+c e vamos provar que $x = \frac{-b \pm \sqrt{\Delta}}{2a}$

ax^2+bx+c * a está multiplicando x^2, vamos dividir por a

$$x^2+\frac{b}{a}x+\frac{c}{a} \to x^2+\frac{b}{a}x+\left(\frac{b}{2a}\right)^2+\frac{c}{a}=\left(\frac{b}{2a}\right)^2 \to \left(x+\frac{b}{2a}\right)^2=\left(\frac{b}{2a}\right)^2-\frac{c}{a} \to \mathrm{I}$$

$$\mathrm{I}\quad \left(x+\frac{b}{2a}\right)^2=\left(\frac{b}{2a}\right)^2-\frac{c}{a} \Rightarrow \left(x+\frac{b}{2a}\right)^2=\frac{b^2}{4a^2}-\frac{c}{a} \Rightarrow \left(x+\frac{b}{2a}\right)^2=\frac{b^2-4ac}{4a^2} \quad \mathrm{II}$$

$$\mathrm{II}\quad \left(x+\frac{b}{2a}\right)^2=\frac{b^2-4ac}{4a^2} \Rightarrow \left(x+\frac{b}{2a}\right)=\sqrt{\frac{b^2-4ac}{4a^2}} \Rightarrow x+\frac{b}{2a}=\frac{\sqrt{\Delta}}{2a} \quad \mathrm{III}$$

$$\mathrm{III}\quad x+\frac{b}{2a}=\frac{\sqrt{\Delta}}{2a} \to x=\frac{-b}{2a}\pm\frac{\sqrt{\Delta}}{2a} \Rightarrow x=\frac{-b\pm\sqrt{\Delta}}{2a} \quad //$$

$$*\ \left(x+\frac{b}{2a}\right)^2 = x^2+2x.\frac{b}{2a}+\left(\frac{b}{2a}\right)^2$$

$$*\ b^2-4ac=\Delta$$

PRODUTOS NOTÁVEIS IMPORTANTES

$$(a+b)^2 = a^2+2.ab+b^2$$

$$(a-b)^2 = a^2-2ab+b^2$$

$$(a+b).(a-b) = a^2-b^2$$

$$(a+b)^3 = a^3-3a^2b+3ab^2+b^2$$

$$(a-b)^3 = a^3+3a^2b+3ab-b^2$$

$$a^3-b^3 = (a-b)(a^2+ab+b^2)$$

$$a^3+b^3 = (a+b)(a^2-ab+b^2)$$

$$a+b+c)^3 = a^3+b^3+c^3+2ab+2ac+2bc$$